6.7 Summary

- In the long line of paradigms in the evolution of programming, from structured to functional to object-oriented to concurrent and parallel programming, blockchain programming is emerging as the next significant component of modern systems.
- The concept of on-chain and off-chain data and operations distinguishes blockchain programming from traditional programming.
- More than transactions are stored on-chain; other data and hashes are stored on-chain as well to support the robustness and usability of the blockchain. These records include the state and state transitions of variables in smart contracts, function return values, events emitted, and items such as logs, all of which are stored in the header of the blockchain to provide provenance for the existence of on-chain data of a specific value.
- Off-chain data sources are application-dependent and are free from the limitations of the blockchain.
- Blockchain enables you to record events emitted by smart contract functions. This feature provides a way other than the function return value to provide notifications from the blockchain layer to upper-level applications.
- Auctions and marketplace models are well suited for blockchain applications.
- End-to-end Dapp design and development roadmap were illustrated by two use cases: blind auction and ASK airline.
- The blind auction Dapp illustrated the use of two types of on-chain data: receipts and event logs.